A Falsified Bar-Instability Criterion Reveals an Independent Disk-State Parameter from EFC-Modified Disk Kinematics

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2026-04-25 **Stage-1 candidate** v0.1

Stage-1 candidate result. Submitted as preregistration of a disk-state observable; ***not*** *a validated bar-onset criterion.* *DOI (Figshare): 10.6084/m9.figshare.PENDING*

Atlas registration: Phenomenon `disk_evolutionary_state`; AtlasPredictions `EFC_disk_state_logSigma_correlation_v0.1` and `EFC_disk_state_fgas_correlation_v0.1`; both with `test_status = candidate_external_support`.

Abstract

We test the prediction that an EFC-modified Toomre-Q stability parameter

$$\Pi_{\rm EFC}(R) \equiv \frac{\sigma_R(R)\,\kappa_{\rm eff}(R)}{3.36\,G\,\Sigma(R)}, \qquad \kappa_{\rm eff}^2 = \kappa^2 + \zeta\,\frac{g_{\rm EFC}}{R}, \qquad g_{\rm EFC} = \alpha\,v\,R\,\frac{d\Omega}{dR} \tag{1}$$

predicts bar morphology in galactic disks. On a sample of $N = 43$ SPARC galaxies with morphological labels obtained from SIMBAD (15 SB + 28 SA, dropping SAB and irregular morphologies), the criterion fails: AUC $= 0.22$ in the local formulation (v0.1), AUC $= 0.31$ in a global resonance-aware formulation (v0.2 with inner Lindblad resonance, corotation, swing-X, and a peaked window function). In both cases the model ranks anti-correlated with bar truth in the physically allowed ζ regime ($\zeta_{\max} \approx 1.52$ from the $\kappa_{\rm eff}^2 \geq 0$ constraint), and the expected per-regime monotonicity (FLOW > TRANSITION > LATENT) breaks at $N = 43$. The bar-prediction track is closed.

In the same data, however, $\Pi_{\min}$ in the disk window correlates strongly with disk structural state. On SPARC $N = 112$ we find Spearman $\rho(\Pi_{\min}, \log\Sigma_{\rm disk}) = -0.71$ and $\rho(\Pi_{\min}, f_{\rm gas}) = +0.65$. The signal survives random-half splits (3 seeds, all 6 sub-samples consistent), substitution of the V-decomposition-based $f_{\rm gas}$ with an independent $M_{\rm HI}/(M_{\rm HI}+M_*)$ estimate ($\rho = +0.53$), stratification by original rotation-curve reduction pipeline (3 sources, all consistent), and replication on the xGASS catalog of 98 SDSS+Arecibo galaxies completely independent of SPARC: $\rho(\Pi_{\rm proxy}, \log\Sigma_*) = -0.49$, $\rho(\Pi_{\rm proxy}, f_{\rm gas}) = +0.67$. First-order partial Spearman tests (§4.6) show that Π is not reducible to $\log\Sigma$ alone (residual $\rho(\Pi, \log\Sigma | f_{\rm gas}) \neq 0$ in both samples) but the partition between $\log\Sigma$ and $f_{\rm gas}$ as predictors is sample-dependent.

We therefore reject EFC-BIC as a bar-instability predictor and instead identify Π as a candidate disk-state proxy defined on a coupled $(\Sigma, f_{\rm gas})$ manifold, rather than as a separable one-parameter observable. We register $\Pi_{\min}$ (and its scalar $\Pi_{\rm proxy}$) as a Stage-1 candidate disk evolutionary state parameter on this joint manifold, with negative projection onto stellar surface density and positive projection onto gas-mass fraction in late-type rotation-supported disks at $z \approx 0$.

1 Problem and Motivation

Bar morphology in galactic disks is a documented blind spot in cosmological theory. A graph traversal of the EFC Framework Atlas confirms that none of 16+ frameworks at galactic-scale L3 (ΛCDM, MOND, Verlinde 2010/2016, $f(R)$ Starobinsky, Horndeski, Kaniadakis HDE, Jacobson Thermodynamic, Padmanabhan Entropic, IEG, Modified Entropic Odintsov, EDE, MiHsC, Negentropic Gravity, Mass-to-Horizon variants, EFC) registers an explicit AtlasPrediction for bar instability. Bar formation is sekulær and history-dependent (gas accretion, tidal triggers, prior disk instabilities), and the standard Toomre-Q parameter — local axisymmetric instability — does not by construction couple radii or set a global $m = 2$ mode amplification.

The EFC Bar Instability Criterion (EFC-BIC) was registered in this Atlas (validation candidate `vc_2f367f3bfc86`) as the first attempt by any framework in the Atlas to fill this gap. The criterion modifies the epicyclic frequency by an entropy-flow term:

$$\kappa_{\text{eff}}^2 = \kappa^2 + \zeta\, g_{\text{EFC}}/R \tag{2}$$

with ζ a universal dimensionless coupling and $g_{\text{EFC}} = \alpha\, \nabla S$ an EFC-derived effective acceleration. The original specification used a longitudinal proxy $\nabla S \equiv d(v^2)/dr$, predicting $\Pi_{\text{EFC}} < 1$ as the bar-onset condition. This paper reports the empirical test of that prediction.

2 Pipeline

The complete pipeline lives in `efc_bic_pipeline.py`, `sparc_loader.py`, and ζ-scan + diagnostic scripts in the supplementary repository. Key fixed choices (same across every reported number):

- **Derivative method:** central differencing with parallel Savitzky–Golay (window=5, order=2). A scipy `savgol_filter` numerical-stability bug at large physical Δx ($\sim 10^{19}$ m) was identified and worked around by computing SG on dimensionless indices and rescaling — without this fix Π_{SG} flips sign relative to Π_{raw}.
- **σ_R prescription:** Jeans-inspired $\sigma_R(R) = \sqrt{\pi G \Sigma(R) R_d}$, with constant 30 km/s as fallback only. The constant fallback was tested and shown to produce unphysical Π in low-Σ dwarfs (range up to 1500); Σ-based σ_R is held as primary throughout.
- **R_d:** independent fit of $V(R) = V_{\text{flat}} \tanh(R/R_d)$ to each SPARC rotation curve, not the SPARC-175 EFC-fit value (which uses a different model parameterization).
- **α:** taken as $1 - L$ per galaxy from `sparc175_classified.json`, where L is the EFC latency fraction. Bounded to $[0.05, 0.95]$.
- **R_{eval}:** $2.2\, R_d$ primary; median(r) fallback if outside data range.
- **Π_{min}:** minimum over $R \in [R_d, 3R_d]$ (the disk window).
- **ζ:** 1.0 unless explicitly varied.

A `proxy_mode` parameter governs the entropy-gradient form: `v_squared` for the original $\nabla S = d(v^2)/dr$, `shear` for $\nabla S = v\, r\, d\Omega/dr$ (introduced in §3.2). Default `v_squared` preserves the v0.1 specification; the shear form is dimensionally equivalent and used in v0.2 onward.

3 Bar-Prediction Falsification

3.1 v0.1 with $d(v^2)/dr$ proxy

Synthetic verification (3 tanh rotation curves, NGC 6503, NGC 3198, NGC 2841 templates) showed Π in a stable range (3.3–5.9) with derivative consistency tolerance 0.2%. RAR-degeneracy test on the synthetic sample yielded $R^2 = 0.97$ (degenerate, as expected for analytical tanh curves).

On the first SPARC pilot ($N = 9$, 3 per regime: FLOW DDO 154/DDO 161/DDO 168; TRANSITION D631-7/DDO 064/F 563-1; LATENT ESO 563-G021/IC 4202/NGC 0247):

- $\Pi_{\min}$ (Jeans σ_R) median $= 8.57$, range $[1.73, 53.91]$
- RAR-degeneracy $R^2 = 0.211 \rightarrow$ `independent_signal` (NOT degenerate with $g_{\rm obs}$ — the largest pre-test risk eliminated)
- σ-sensitivity test: 22% binary classification flip when σ_R was changed from Jeans to constant — the flips concentrate in LATENT regime

A diagnostic of the EFC-term magnitude $|EFC|/|\kappa^2|$ found median ratio 0.117 in the disk window, max 0.49 — substantial, not negligible. **But signed:** 9/9 galaxies showed POSITIVE EFC$/\kappa^2$ in the bar-formation zone (rising rotation curves $\rightarrow \nabla S(v^2) > 0$). With $\alpha > 0$ and $\zeta > 0$, this means the EFC term *stabilises* the disk where bars should form. The criterion is structurally incapable of predicting bar onset in this formulation.

3.2 v0.1 with shear proxy

Replacing the entropy-gradient proxy with $\nabla S = v\, r\, (d\Omega/dr)$ — dimensionally equivalent, using the differential rotation rate that quantifies orbit phase decoherence — gives the correct sign:

- 21/21 galaxies in the $N = 21$ expanded pilot had $\nabla S < 0$ in the disk $\rightarrow g_{\rm EFC} < 0 \rightarrow \kappa^2_{\rm eff} < \kappa^2 \rightarrow$ reduces Π toward instability
- Per-regime medians at $\zeta = 1$: FLOW 7.14, TRANSITION 5.43, LATENT 1.87 — apparent monotonicity (LATENT closest to threshold)
- ζ-grid scan: median $\Pi_{\min} < 1$ crosses at $\zeta \approx 3$ for all regimes; LATENT remains physical ($\kappa^2_{\rm eff} > 0$) up to $\zeta \approx 5$
- RAR-degeneracy $R^2 = 0.092 \rightarrow$ still `independent_signal`

This is a real structural fix. But the apparent $N = 21$ monotonicity proved to be a small-sample artefact.

3.3 Classification fit (v0.1 shear, $N = 43$ cleanly labeled)

Bar/no_bar labels were obtained from SIMBAD (`Morphological type` field, parsed with a strict de Vaucouleurs bar-code classifier). Classes: 15 SB (bar), 28 SA (no_bar), 31 SAB (mixed — dropped), 14 not-disk (irregulars/early — dropped), 79 unknown (T-type only or ambiguous — dropped).

Result: **AUC** $= 0.217$ **at** $\zeta = 1.0$ **in the physically allowed regime (**$\zeta_{\rm max\ physical} = 1.52$ **from worst-case** $\kappa^2_{\rm eff} \geq 0$**).**

Per-regime medians at $\zeta = 1$ broke the apparent ordering: FLOW 1.81, TRANSITION 3.21, LATENT 1.14 — TRANSITION highest, not lowest. The $N = 21$ ordering had been a coincidence of which galaxies were alphabetically first per regime.

Misclassification pattern:

- **False negatives** (SB predicted no_bar): NGC 2903 (SBbc, $\Pi = 1.34$), ESO 116-G012 (SBd, $\Pi = 3.72$), UGC 05986 (SB, $\Pi = 4.22$), and 6 more — real bars sit at high Π.
- **False positives** (SA predicted bar): NGC 2955, UGC 06786, UGC 06787 — LSB galaxies with low $\Sigma \to$ low Π via low Σ, not via EFC effect.

The model detects "low surface density" not "barred". AUC < 0.5 means ranking is anti-correlated with bar truth.

3.4 v0.2 resonance-aware formulation

To address the local-vs-global concern (bar formation is a global $m = 2$ mode coupling), we built a resonance-aware criterion: at each trial corotation radius r_{CR} (scanned over $r_{\mathrm{CR}}/R_d \in [1, 4]$) compute pattern speed $\Omega_p \equiv \Omega(r_{\mathrm{CR}})$, find the inner Lindblad resonance r_{ILR} via $\Omega - \kappa_{\mathrm{eff}}/2 = \Omega_p$, compute the Toomre-$X$ swing-amplification parameter $X = \kappa_{\mathrm{eff}}^2 \cdot r_{\mathrm{CR}}/(2\pi G \Sigma m = 2)$, apply a Gaussian window $A(X)$ peaked at $X = 1.5$, and define bar-score $= A(X)/\max(Q_{\mathrm{ILR}}, \epsilon)$ with Q_{ILR} computed at r_{ILR}.

Result: **AUC** $= 0.310$ **at best** $\zeta = 1.5$**.** Recall $= 18\%$ (2/11 bars). The score is dominated by ILR-absent vs ILR-present, which is itself dominated by data-window geometry rather than bar physics. 79% of galaxies have ILR absent in the data window. Real SBs with ILR present (e.g., UGC 03546 at $Q_{\mathrm{ILR}} = 2.08$) get score ≈ 0; real SBs with high X (e.g., NGC 3109 SB(s)m at $X = 48$) fall outside the $A(X)$ window.

Verdict: bar-prediction track CLOSED. Two independent formulations (local v0.1 and global v0.2), both with correctly-implemented physics, both fail in the same direction with AUC < 0.5 in the physically allowed parameter regime. The model ranks anti-correlated with bar morphology. This is interpreted as an information-limit result: snapshot disk kinematics — even with EFC modification of the epicyclic frequency and a global $m = 2$ resonance treatment — does not contain enough information to predict bar status. Bar formation depends on history (gas accretion, tidal triggers, prior instabilities) that current state cannot recover.

The Atlas Phenomenon `Bar Instability in Galactic Disks` and linked EFCValidation are flagged FALSIFIED 2026-04-25 with full evidence pointers.

4 Pivot to Disk-State Parameter

The same $\Pi_{\min}$ that fails at bar prediction has, in the same data, consistent monotonic correlations with disk structural observables. We did not adjust the pipeline; we adjusted the question.

4.1 Within-SPARC correlations ($N = 112$)

Observable	Spearman $\rho(\Pi_{\min}, \cdot)$	t-stat
$\log \Sigma_{\rm disk}$ (median in $[R_d, 3R_d]$)	$\mathbf{-0.714}$	-10.7
$f_{\rm gas}$ ($V^2_{\rm gas}/V^2_{\rm baryon}$ median)	$\mathbf{+0.648}$	$+8.9$
disk dominance ($V^2_{\rm disk}/V^2_{\rm obs}$)	-0.409	-4.7
$V_{\max}$	-0.461	-5.6

Table 1: SPARC $N = 112$ Spearman correlations.

Regime	N	$\Pi_{\min}$	$\log \Sigma$	$f_{\rm gas}$	disk_dom
FLOW	47	2.14	+1.30	0.06	0.53
TRANSITION	44	3.98	+0.44	0.18	0.22
LATENT	21	**1.52**	**+2.17**	**0.00**	0.41

Table 2: Per-regime medians at $\zeta = 1$.

LATENT galaxies (well-known massive spirals: NGC 2841, NGC 2998) sit at LOW Π / HIGH Σ / ZERO gas — the "mature, structurally settled" corner. TRANSITION galaxies sit at HIGH Π / LOW Σ / HIGH gas — the "diffuse, kinematically immature" corner.

Binary-prediction AUCs (treating $\Pi_{\min}$ as score, predicting tertile membership of independent observables):

- gas-rich (top tertile of $f_{\rm gas}$): AUC = **0.812**
- low-Σ (bottom tertile of $\log \Sigma$): AUC = **0.814**
- disk-faint (bottom tertile of $V^2_{\rm disk}/V^2_{\rm obs}$): AUC = 0.663

4.2 Robustness: random splits (Stage D)

We split $N = 112$ randomly with three seeds (42, 7, 13), each into halves of 56. All six sub-samples preserved both signs:

seed	half	N	$\rho(\log \Sigma)$	$\rho(f_{\rm gas})$
42	1	56	−0.755	+0.698
42	2	56	−0.671	+0.583
7	1	56	−0.826	+0.745
7	2	56	−0.594	+0.526
13	1	56	−0.759	+0.614
13	2	56	−0.689	+0.680

Table 3: SPARC random half-split, ρ-range: $\log \Sigma \in [-0.83, -0.59]$, $f_{\rm gas} \in [+0.53, +0.75]$. Sign held in all 6 splits.

4.3 Robustness: observable substitution (Stage C)

Internal $f_{\rm gas}$ ($V_{\rm gas}^2/V_{\rm baryon}^2$ from rotation-curve component decomposition) and Π share the same kinematic source. Substituting $f_{\rm gas}$ with an independent estimate from the SPARC table1 master catalog ($M_{\rm HI}$ from radio-survey HI integration, $M_* = \Upsilon \cdot L[3.6]$ from Spitzer photometry, $\Upsilon_{\rm disk} = 0.5\, M_\odot/L_\odot$, $f_{\rm gas,ext} = 1.4M_{\rm HI}/(1.4M_{\rm HI} + M_*)$):

Mode	N	$\rho(\Pi, \log\Sigma)$	$\rho(\Pi, f_{\rm gas})$
Internal ($V_{\rm gas}$-based)	112	-0.714	$+0.648$
External ($M_{\rm HI}/L_{3.6}$)	112	-0.714	$\mathbf{+0.530}$

Table 4: Stage C: cross-check $\rho(\text{internal}, \text{external})\ f_{\rm gas} = +0.84$. Sign holds; strength reduced by 18%.

4.4 Robustness: reduction-pipeline stratification (Stage B-prime)

SPARC table1's `Ref` column identifies the original rotation-curve reduction. Within $N = 112$, three reductions have $N \geq 10$ (different research groups, telescopes, pipelines):

Source	Author	N	$\rho(\log\Sigma)$	$\rho(f_{\rm gas})$
VS01	Verheijen & Sancisi 2001	17	$\mathbf{-0.914}$	$\mathbf{+0.816}$
Sw09	Swaters 2009	15	-0.704	$+0.543$
No07	Noordermeer 2007	11	$\mathbf{-0.900}$	$\mathbf{+0.727}$

Table 5: Stage B-prime: all three reductions preserve both signs and $|\rho| > 0.5$. Pipeline-reduction artefact excluded.

4.5 External sample validation (Stage A): xGASS

The decisive test is sample independence. Public THINGS / LITTLE THINGS catalogs at VizieR do not expose the $V_{\rm gas}/V_{\rm disk}$ component decomposition our pipeline requires (only total scaled rotation curves). We instead used xGASS (Catinella et al. 2018) — a stellar-mass-selected sample of 1179 galaxies at $z \approx 0.01 - 0.05$ with SDSS Petrosian photometry ($\log\Sigma_*$, R_e), Arecibo HI line widths (W_{50}), and HI masses. After cross-matching tablea1 + tablea2, requiring inclination $> 30°$ and HI quality flag ≤ 2, we have $N = 98$ galaxies independent of SPARC at the level of sample, observable derivation, and reduction pipeline.

For xGASS we evaluated a scalar $\Pi_{\rm proxy}$ at R_e:

$$\Pi_{\rm proxy}(R_e) = \frac{V_{\rm rot} \cdot \sigma_{\rm Jeans}}{3.36\, G\, \Sigma_*\, R_e}, \quad V_{\rm rot} = \frac{W_{50,c}}{2 \sin i}, \quad \sigma_{\rm Jeans} = \sqrt{\pi G \Sigma_* R_e} \tag{3}$$

(Same Jeans σ_R formula as the SPARC pipeline; only the spatial evaluation choice differs — scalar at R_e instead of radial-profile minimum over $[R_d, 3R_d]$.) External $f_{\rm gas}$: $M_{\rm HI}/(M_{\rm HI}+M_*)$ directly from xGASS tablea2.

4.6 Partial-correlation analysis and degeneracy structure

We tested whether the Π correlations with stellar surface density and gas fraction represent independent signals or a reparameterization of a single underlying variable.

In SPARC ($N = 112$), $\log\Sigma$ and $f_{\rm gas}$ are strongly anti-correlated ($\rho = -0.90$). Partial Spearman analysis yields $\rho(\Pi, \log\Sigma \,|\, f_{\rm gas}) = -0.394$, while $\rho(\Pi, f_{\rm gas} \,|\, \log\Sigma) = +0.017$. In

Sample	N	$\rho(\Pi, \log \Sigma_*)$	$\rho(\Pi, f_{\mathrm{gas}})$	sign
SPARC (reference)	112	-0.71	$+0.65$	$-\,+$
xGASS (Stage A)	**98**	$\mathbf{-0.49}$	$\mathbf{+0.67}$	$\mathbf{-\,+}$

Table 6: Both signs preserved. $|\rho| \geq 0.49$. p-values $< 10^{-7}$. The signal is not confined to SPARC.

this sample, the Π–f_{gas} correlation is therefore largely explained by its mutual correlation with Σ.

In xGASS ($N = 98$), the predictor correlation is weaker ($\rho = -0.51$). Here we find $\rho(\Pi, \log \Sigma \mid f_{\mathrm{gas}}) = -0.232$ and $\rho(\Pi, f_{\mathrm{gas}} \mid \log \Sigma) = +0.554$, indicating a substantial residual dependence on gas fraction after controlling for Σ.

Sample	$\rho(\log \Sigma, f_{\mathrm{gas}})$	$\rho(\Pi, \log \Sigma \mid f_{\mathrm{gas}})$	$\rho(\Pi, f_{\mathrm{gas}} \mid \log \Sigma)$
SPARC ($N = 112$)	$\mathbf{-0.901}$	$\mathbf{-0.394}$	$\mathbf{+0.017}$
xGASS ($N = 98$)	$\mathbf{-0.511}$	$\mathbf{-0.232}$	$\mathbf{+0.554}$

Table 7: First-order partial Spearman: the decomposition into "Σ-driven" vs "f_{gas}-driven" is sample-dependent.

This asymmetry shows that the decomposition into "Σ-driven" versus "f_{gas}-driven" contributions is sample-dependent. We therefore do not interpret Π as independently measuring either quantity. Instead, Π appears to trace a joint disk-state manifold defined by the coupled evolution of stellar surface density and gas fraction. Accordingly, all claims in this work are restricted to monotonic relations within this joint space, rather than separable one-parameter dependencies. This is also reflected in the Atlas registration: the `known_degeneracy` field on Phenomenon `disk_evolutionary_state` records this explicitly.

5 Results and Interpretation

5.1 Empirical result

Across two independent samples (SPARC $N = 112$, xGASS $N = 98$) and four independent robustness tests (random sub-samples, observable substitution, reduction-pipeline stratification, sample replication), an EFC-modified Toomre-Q-style parameter Π (or its scalar proxy at R_e) shows consistent monotonic correlations with disk structural observables:

- $\rho(\Pi, \log \Sigma_*)$: $\mathbf{-0.71}$ (SPARC) and $\mathbf{-0.49}$ (xGASS)
- $\rho(\Pi, f_{\text{gas}})$: $\mathbf{+0.65}$ (SPARC) and $\mathbf{+0.67}$ (xGASS)
- $|\rho|$ ranges 0.49–0.83 across all sub-tests; sign preserved without exception
- Partial-correlation analysis (§4.6) confirms the signal is not reducible to $\log \Sigma$ alone

5.2 Interpretation as disk-state proxy

The empirical pattern is consistent with Π capturing a property of the disk's dynamical-structural state rather than its mode-coupling instability spectrum: Π is small for compact, high-Σ, gas-poor, dynamically settled late-type disks (the Atlas `LATENT` regime, populated by massive evolved spirals such as NGC 2841 and NGC 2998); Π is large for diffuse, low-Σ, gas-rich, kinematically less-evolved disks (Atlas `TRANSITION` regime, populated by low-surface-brightness gas-rich systems). The correlation patterns survive across measurement modes (radial-profile minimum vs. scalar at R_e), gas-fraction definitions (V-decomposition vs. M_{HI}/M_*), reduction pipelines, and sample selections.

We propose Π as a Stage-1 candidate disk-state proxy in the joint (Σ, f_{gas}) plane, and we explicitly do *not* propose it as: (i) a bar-instability criterion, (ii) an independent measurement of either Σ_* or f_{gas} separately, (iii) a fundamental constant, or (iv) a global cosmological observable beyond the validity scope (late-type rotation-supported disks at $z \approx 0$).

6 Limitations

1. **Π and Π_{proxy} are not identical.** The SPARC pipeline computes Π_{min} over the disk window $[R_d, 3R_d]$ from a full radial profile. The xGASS test uses a scalar evaluation at R_e because radial profiles are not in the public catalog. Both use the same Σ-Jeans σ_R formula and same shear-based g_{EFC}, but the spatial reduction differs.
2. **Mutual correlation of predictors.** $\log \Sigma_*$ and f_{gas} are themselves anti-correlated in the local universe (massive disks tend to be gas-poor). Multivariate decomposition does not cleanly separate Σ-driven from f_{gas}-driven contribution. Quantified explicitly in §4.6.
3. **No IFU-resolved test.** A radial-profile validation on IFU data (MaNGA, SAMI, CALIFA) is the next-priority extension.
4. **No stage-3 environmental test.** Sample ranges over $z \approx 0 - 0.05$, dominated by isolated disks. Cluster, group, and high-z behaviour unconstrained.
5. **ζ uncalibrated as a fundamental coupling.** $\zeta = 1.0$ held fixed throughout. Rank correlations are weakly dependent on $\zeta \in [0.5, 1.5]$ but this paper does not constrain ζ as a universal physical constant.

7 Predictions and falsifiable next tests

We register the following predictions as Atlas candidates with `test_status = candidate_external_support`:

P1 + P2 — Joint manifold projection (registered): $\Pi_{\rm min}$ (or $\Pi_{\rm proxy}$) varies monotonically across the joint $(\log \Sigma_*, f_{\rm gas})$ space of late-type rotation-supported disks at $z \approx 0$, with negative projection onto $\log \Sigma_*$ and positive projection onto $f_{\rm gas}$. Marginal Spearman correlations on any SPARC/xGASS-comparable sample shall preserve both signs with $|\rho| \geq 0.4$. AtlasPredictions: `EFC_disk_state_logSigma_correlation_v0.1` (value $= -0.60 \pm 0.12$) and `EFC_disk_state_fgas_correlation_v0.1` (value $= +0.66 \pm 0.10$). The two predictions are NOT independent — they project a single underlying joint-manifold relation onto two correlated coordinate axes (see §4.6).

P3 — IFU radial test (predicted, not yet tested): On MaNGA / SAMI / CALIFA spatially-resolved sub-samples with $N \geq 30$ late-type disks, the radial $\Pi(R)$ profile shall reach its minimum within the disk window $[R_e, 3R_e]$, and $\Pi_{\rm min}$ shall preserve P1 and P2 with sign and $|\rho| \geq 0.4$.

P4 — Cosmological-zoom prediction (predicted, not yet tested): In FIRE-2 / IllustrisTNG cosmological-zoom simulations, the same galaxy at successive redshifts shall show $\Pi_{\rm min}$ decreasing as Σ_* grows and $f_{\rm gas}$ declines.

P5 — Counter-example falsifiability: A galaxy with high $\Pi_{\rm min}$ and high Σ_* (top tertile in both) or with low $\Pi_{\rm min}$ and low Σ_* (bottom tertile in both), at $> 2\sigma$ from the SPARC+xGASS regression, would be a falsification candidate.

8 Atlas registration (for traceability)

- **Phenomenon:** `Disk Evolutionary State` (canonical_id `disk_evolutionary_state`). `observable_type = joint_manifold_projection`, regime L3, `validity_scope`, `known_degeneracy`, and `manifold_axes = [log_Sigma_star, f_gas]` fields explicitly populated.
- **AtlasPrediction P1:** `EFC_disk_state_logSigma_correlation_v0.1`, value -0.60, units `Spearman_rho_Pi_vs_log_Sigma_disk`, `observable_definition` records both SPARC radial-profile and xGASS scalar-at-R_e modes. `test_status = candidate_external_support`.
- **AtlasPrediction P2:** `EFC_disk_state_fgas_correlation_v0.1`, value $+0.66$, units `Spearman_rho_Pi_vs_f_gas`, `observable_definition` records both $V_{\rm gas}$-based and $M_{\rm HI}$-based $f_{\rm gas}$. `test_status = candidate_external_support`.
- **Bar-instability node UNCHANGED:** `Phenomenon: Bar Instability in Galactic Disks` and linked `EFCValidation` retain `pipeline_status = FALSIFIED 2026-04-25` and `status = FALSIFIED`. No back-linking from disk-state to bar.

9 What this paper is not

It is not a confirmation that EFC is correct. It is not a measurement of ζ. It is not a fundamental law. It is not a bar-instability criterion (which was tried and failed). It is a Stage-1 candidate observable with cross-dataset support, registered transparently to allow independent groups to test, refute, or confirm.

Reproducibility

All code, data, and outputs are in the supplementary repository (`scripts/efc_bic_pilot/`):

- Pipeline: `efc_bic_pipeline.py`, `sparc_loader.py`
- Bar-prediction failure: `classify_fit.py`, `efc_bic_v02_resonance.py`
- Disk-state pivot: `disk_state_analysis.py`
- Robustness: `test_DC_validation.py`, `test_A_xgass.py`
- Outputs: `output/disk_state.json`, `output/test_DC.json`, `output/test_A_xgass.json`, `output/classify_fit.json`, `output/v02_resonance.json`
- Bar labels: `fetch_bar_labels.py`, `data/sparc/sparc_bar_labels.json`
- xGASS data: `data/external/xgass/`

References

1. Lelli, F., McGaugh, S. S., & Schombert, J. M. 2016, AJ, 152, 157 (SPARC). 10.3847/0004-6256/152/6/157
2. Catinella, B., et al. 2018, MNRAS, 476, 875 (xGASS). 10.1093/mnras/sty089
3. Walter, F., et al. 2008, AJ, 136, 2563 (THINGS). 10.1088/0004-6256/136/6/2563
4. Hunter, D. A., et al. 2012, AJ, 144, 134 (LITTLE THINGS). 10.1088/0004-6256/144/5/134
5. Buta, R. J., et al. 2015, ApJS, 217, 32 (S4G morphology). 10.1088/0067-0049/217/2/32
6. de Vaucouleurs, G., et al. 1991, *Third Reference Catalogue of Bright Galaxies* (RC3, Springer) — no DOI.
7. Wenger, M., et al. 2000, A&AS, 143, 9 (SIMBAD). 10.1051/aas:2000332
8. Toomre, A. 1964, ApJ, 139, 1217 (Q parameter). 10.1086/147861
9. Toomre, A. 1981, in *The Structure and Evolution of Normal Galaxies*, ed. S. M. Fall & D. Lynden-Bell (Cambridge UP), p. 111 (swing amplification) — no DOI.
10. Sellwood, J. A., & Wilkinson, A. 1993, Rep. Prog. Phys., 56, 173 (bar instability theory). 10.1088/0034-4885/56/2/001